

James Ebbah-Idala

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JUNIOR PENETRATION TESTER

Early-career cybersecurity professional focused on **application security, vulnerability assessment, penetration testing, and offensive security**. Hands-on experience building and testing security-focused applications, analyzing vulnerable web applications through CTF and security lab environments, and developing Python-based tooling. Comfortable reading code and understanding application behavior, authentication flows, attack surfaces, and potential security weaknesses. **CompTIA Security+ certified** with practical experience across Hack The Box, TryHackMe, CTF competitions, and cybersecurity challenge development.

TECHNICAL SKILLS

Application Security: Vulnerability Assessment, Web Application Security, API Security, OWASP Top 10, Authentication & Authorization, Injection, Access Control, Input Validation, Information Disclosure, Secure Coding, Security Testing, Vulnerability Validation

Programming: Python, Flask, SQL, JavaScript, HTML/CSS, Bash

Security Tools: Burp Suite, Nmap, Wireshark, Metasploit, Kali Linux, Linux, Splunk, Wazuh, Git/GitHub, Hack The Box, TryHackMe

Development: Git, GitHub, REST APIs, SQLite, PyCryptodome, Requests, Python Automation

Security Practices: Penetration Testing, CTFs, Vulnerability Triage, Security Documentation, Attack-Surface Analysis, Remediation Analysis, Risk Assessment

CYBERSECURITY EXPERIENCE

Hackfest Québec — CTF Challenge Designer / Cybersecurity Contributor

2026 – Present

- Currently collaborating with the Hackfest cybersecurity team to **design and develop challenges** for an upcoming cybersecurity competition.
- Apply offensive-security concepts to create realistic scenarios involving **vulnerability discovery, exploitation, application behavior, and attack paths**.
- Test challenges from an attacker perspective to identify unintended solutions, implementation issues, and weaknesses in challenge infrastructure.
- Collaborate with other security practitioners to refine challenge concepts, validate solutions, and improve technical quality and participant experience.
- Develop practical understanding of how vulnerabilities can be intentionally introduced, discovered, validated, and explained in realistic security scenarios.

Cybersecurity CTFs & Security Labs — Hack The Box / TryHackMe

- Conduct hands-on **vulnerability assessment and penetration testing** against intentionally vulnerable web applications, services, and infrastructure.
- Identify and validate common application and infrastructure weaknesses including **injection, authentication flaws, access-control issues, information disclosure, misconfigurations, and privilege escalation**.
- Use **Burp Suite, Nmap, Linux/Kali, browser developer tools, and custom Python scripts** to enumerate attack surfaces and investigate application behavior.
- Analyze HTTP requests, parameters, responses, authentication mechanisms, and application logic to understand the root cause and exploitability of vulnerabilities.
- Document attack paths, technical findings, evidence, impact, exploitation methodology, and remediation considerations through security writeups.
- Collaborate in CTF environments to investigate unfamiliar technologies, develop attack hypotheses, validate findings, and communicate technical discoveries.

BSides — Hack The Box CTF Competitor

2026

- **Placed 2nd in a BSides Hack The Box CTF** as part of a competitive cybersecurity team, solving hands-on challenges across web security, networking, privilege escalation, and other offensive-security domains.
- Collaborated with teammates to enumerate targets, identify vulnerabilities, develop exploitation paths, and validate findings under time constraints.
- Applied practical offensive-security techniques using **Burp Suite, Nmap, Linux/Kali, and custom scripts** to investigate and exploit security weaknesses.
- Analyzed application behavior, HTTP requests, services, and attack surfaces to develop and validate potential exploitation paths.
- Communicated technical findings and attack paths with teammates to prioritize promising avenues and efficiently solve challenges.

PROJECTS

Secure Password Manager

Python, Flask, SQLite, PyCryptodome

- Developed a security-focused **Flask web application** for managing credentials using encryption and secure data-handling practices.
- Implemented **AES encryption with PyCryptodome** to protect stored sensitive credentials and integrated SQLite for application data storage.
- Integrated the **Have I Been Pwned API** using the k-anonymity model to check password compromise without transmitting plaintext passwords.
- Analyzed application attack surfaces including authentication, credential handling, input processing, API interaction, and sensitive-data storage.
- Applied secure-development principles while building and testing application functionality in Python and Flask.

Web Application Security Testing Lab

Burp Suite, OWASP, Python, Linux

- Built hands-on experience identifying and validating **OWASP Top 10-style vulnerabilities** in intentionally vulnerable applications on platforms like HackTheBox, TryHackMe and Portswigger.
- Used Burp Suite to intercept and modify HTTP requests, analyze application responses, test parameters, and investigate authentication and authorization behavior.
- Performed manual testing alongside enumeration and security tooling to distinguish exploitable vulnerabilities from non-impactful findings.
- Documented findings with technical evidence, potential impact, attack paths, and recommended remediation approaches.

Security Monitoring Lab

Splunk, Wazuh, Linux

- Built practical experience analyzing security telemetry and investigating suspicious activity using **Splunk and Wazuh**.
- Analyzed logs and endpoint/network events to understand how security findings can be detected, investigated, and correlated with attacker activity.
- Developed foundational experience connecting vulnerability discovery, exploitation, detection, and remediation across security workflows.

EDUCATION

Bachelors Of Science — Computer Science, Cybersecurity Stream
Carleton University

2025
Ottawa, Ontario

CERTIFICATION

CompTIA Security+
HackTheBox Certified Penetration Testing Specialist - In Progress